

JPG File Organizer

Task 3 – Task Automation with Python Scripts

1. Project Overview

The JPG File Organizer is a Python-based automation project that separates JPG and JPEG image files from other file types in a selected source folder. The program automatically creates a destination folder and moves only JPG/JPEG images into it, while leaving files such as PNG, PDF, TXT, MP4 and DOCX in the source folder.

2. Objective

The main objective is to automate a repetitive file-management task. Instead of manually searching for JPG images and moving them one by one, the Python script identifies the required files and organizes them automatically.

3. Technologies Used

Technology / Module	Purpose
Python	Programming language used to develop the automation tool
os	Checks folders, lists files, creates paths and directories
shutil	Moves files from the source folder to the destination folder

4. Project Structure

```
JPG File Organizer/  
■■■ file_organizer.py  
■■■ README.md  
■■■ source_folder/  
    ■■■ image1.jpg  
    ■■■ image2.jpeg  
    ■■■ document.pdf  
    ■■■ image.png
```

The destination_folder is created automatically by the program when it runs.

5. Working Principle

Step	Description
1	The user provides the source folder path.
2	The program checks whether the source folder exists.
3	A destination_folder is created automatically.
4	The program examines each item in the source folder.
5	Only files ending in .jpg, .JPG, .jpeg or .JPEG are selected.
6	Selected image files are moved to destination_folder.
7	Other file types remain in the source folder.
8	The program displays the number of files moved.

6. Example

Before execution:

source_folder: image1.jpg, image2.jpg, document.pdf, notes.txt, image.png

After execution:

source_folder: document.pdf, notes.txt, image.png

destination_folder: image1.jpg, image2.jpg

7. Important Features

- Automatically creates the destination folder.
- Supports both JPG and JPEG extensions.
- Recognizes uppercase and lowercase extensions.
- Leaves non-JPG files untouched.
- Handles duplicate filenames by creating a numbered filename.
- Handles common permission and operating-system errors.
- Shows a completion message and the total number of files moved.

8. Complete Python Code

```
import os
import shutil

def organize_jpg_files(source_folder):
    if not os.path.isdir(source_folder):
        print("\nError: Source folder does not exist.")
        return

    destination_folder = os.path.join(
        os.path.dirname(source_folder),
        "destination_folder"
    )

    os.makedirs(destination_folder, exist_ok=True)

    print("\nSearching for JPG files...")
    print("-----")

    moved_count = 0

    for filename in os.listdir(source_folder):
        source_path = os.path.join(source_folder, filename)

        if not os.path.isfile(source_path):
            continue

        if not filename.lower().endswith((".jpg", ".jpeg")):
            continue

        destination_path = os.path.join(
            destination_folder, filename
        )

        if os.path.exists(destination_path):
```

```

name, extension = os.path.splitext(filename)

counter = 1

while True:

    new_filename = f"{name}_{counter}{extension}"
    new_destination = os.path.join(
        destination_folder, new_filename
    )

    if not os.path.exists(new_destination):
        destination_path = new_destination
        filename = new_filename
        break

    counter += 1

try:

    shutil.move(source_path, destination_path)
    print(f"Moved: {filename}")
    moved_count += 1

except PermissionError:

    print(f"Permission denied: {filename}")

except OSError as error:

    print(f"Could not move {filename}: {error}")

print("-----")
print("TASK COMPLETED")
print("-----")
print(f"JPG/JPEG files moved: {moved_count}")
print(f"Destination: {destination_folder}")
print("-----")

```

```

def main():

    print("=====")

    print("          JPG FILE ORGANIZER")

    print("=====")


    source_folder = input(

        "Enter source folder path: "

    ).strip().strip(' ')


    organize_jpg_files(source_folder)


if __name__ == "__main__":

    main()

```

9. How to Run

1. Open the project folder in VS Code.
2. Open the terminal.
3. Run: **python file_organizer.py**
4. Enter the source folder path, for example: **./source_folder**
5. The program creates destination_folder automatically and moves the JPG/JPEG files.

10. Learning Outcomes

This project provides practical experience with Python functions, loops, conditional statements, directory handling, file paths, the os module, the shutil module, duplicate-file handling and task automation.

11. Conclusion

The JPG File Organizer demonstrates how Python can automate a simple but repetitive computer task. By identifying JPG/JPEG files and separating them from other file types, the project reduces manual effort and keeps folders organized. The solution is lightweight, easy to understand and can be extended later to organize other file types such as PDFs, documents, videos or music.